SAINT MICHAEL COLLEGE OF CARAGA

CAGATCIPHER: AN ENCRYPTION ALGORITHM USING ADVANCED ENCRYPTION STANDARD (AES) AND BLOWFISH WITH COMPLEMENTING KEY MANAGEMENT

A Thesis Proposal Presented To
The Faculty of the College of Computing and Information Sciences

In Partial Requirements for the Degree
BACHELOR OF SCIENCE IN COMPUTER SCIENCE



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March 2026

SAINT MICHAEL COLLEGE OF CARAGA
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ABSTRACT

I. OBJECTIVES

- To effectively determine the optimal performance after implementing the hybrid AES and Blowfish encryption, using speed and responsiveness as key metrics.

- To compare the overall performance of the CAGATCipher compared to other hybrid and standard encryption algorithm.

- To implement effective key management to ensure efficient and secure handling of Encryption within institutional workflows.

II. METHODOLOGY

Developmental-Experimental

This study uses a Developmental-Experimental Research Design, which combines both the creation and assessment of a new hybrid algorithm. To be exact, the study would develop a new algorithm utilizing the implementation of the Advanced Encryption Standard (AES) and Blowfish to create a much more prominent encryption algorithm.

The development phase would involve creating and developing the algorithm, along with the simulated system on which we would test it. On the other hand, the experimental phase would test and evaluate the effectiveness, responsiveness, and security integrity of the encrypted data. These processes will ensure the quality development and integration of the new hybrid algorithm.

III. FINDINGS

IV. RECOMMENDATIONS

V. KEYWORDS: Encryption, Decryption

DEDICATION

This thesis is sincerely dedicated to all those who have helped, inspired, and guided the researchers in their quest to complete this research.

To the researchers' families, whom they love, thank you for your unwavering support, patience, and encouragement during difficulties; your support and faith in them have been their source of inspiration.

To their adviser, Ms. Lealil I. Palacio, your precious insights, careful suggestions, and dedication to excellence have significantly contributed to shaping the path and success of this research.

To their Research 1 teacher, Ms. Daisa O. Gupit, thank you for your compassion and timely assistance that guided us in establishing a firm foundation. The researchers further dedicate this project to the students and faculty of Saint Michael College of Caraga, who enriched it with their inputs and engagement, emphasizing real-world applicability.

For their friends and classmates, moral support, humor, and encouragement helped maintain their footing and focus. You all played a significant role in the process, and the researchers are forever thankful for your presence. None of this could have been done without all of you, and this is a testament to the effort, diligence, and kindness you were so fortunate to receive.

ACKNOWLEDGEMENT

The researchers are deeply grateful to everyone who contributed to the completion of this research project. Their sincerest thanks to the adviser, Ms. Lealil I. Palacio, MSIT (CAR), for her invaluable guidance, encouragement, and support throughout the development of the study.

To their research, one instructor, Mrs. Daisa O. Gupit, MIT, thank you for your patience and understanding during the early stages of this research.

The researchers also extend their appreciation to the panel chairman, Mr. Kenneth Ian B. Barrera, MA, for his constructive feedback and helpful recommendations that contributed to the improvement of this study.

To their families, your love and encouragement kept them going through the journey; to their friends, your support meant the world to them.

The researchers would also like to thank Saint Michael College of Caraga for allowing them to conduct this research within the institution, with special thanks to the dean, Daisa O. Gupit, MIT, Engr. Mirasol S. Rosario, along with the faculty of the research office, participated in the event. Your data gives its true purpose to this system.

To everyone who helped in any way, directly or indirectly, thank you for making this research possible.

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CHAPTER 1

INTRODUCTION

1.1 Project Context

The rapid growth of digital technology has transformed the way data is created, stored, transmitted, and processed across various platforms. As information systems and online communication continue to expand at a remarkable pace, protecting sensitive information has become more crucial than ever. In this context, cryptography plays a vital role in safeguarding private data from unauthorized access, alteration, or misuse. [1]

The Advanced Encryption Standard (AES), originally known as the Rijndael cipher, is currently the world's most widely used symmetric key encryption algorithm [2]. Meanwhile, Twofish—developed by Bruce Schneier and his team as an AES competition finalist—is also a symmetric block cipher capable of using keys up to 256 bits. Although not selected as the official standard, Twofish is recognized for its flexibility, complex key schedule, and high level of security, making it suitable for both hardware and software applications. It is powerful against various cryptanalytic attacks and offers solid performance. With cyberattacks evolving rapidly, combining the best features of multiple algorithms through a hybrid encryption scheme is becoming more important than ever [3]. Using both AES and Twofish together can enhance security and performance, keeping up with growing system complexity [4]. This study focuses on designing and highlighting the value of an AES/Blowfish hybrid encryption approach in today's digital landscape, where trust and reliability are essential.

Around the world, hybrid encryption methods—especially those that combine AES and Blowfish—are being adopted more frequently to address the need for both security and efficiency. Research in cloud computing shows that AES-Blowfish hybrids can deliver better throughput and stronger defenses against man-in-the-middle attacks than using just one algorithm [5]. In the Internet of Things (IoT) field, combining Blowfish with elliptic-curve-based protocols leads to faster processing and lower memory usage [6]. For mobile apps, AES-Blowfish hybrids enhance authentication security without causing noticeable slowdowns [7]. Other studies note that Blowfish performs well in high-throughput scenarios, and hybrid approaches with AES can further improve security and efficiency [8]. AES is particularly good for encrypting large multimedia files, while Blowfish works well for smaller datasets—making a hybrid strategy practical [9].

Recent research has expanded the focus on hybrid cryptographic systems across domains such as cloud computing, healthcare, and big data analytics. Comparative analyses of AES, Blowfish, and Twofish for image encryption have shown that hybrid approaches can optimize both processing speed and security for multimedia applications. [10] In the healthcare sector, hybrid AES-based methods have proven adaptable for protecting sensitive records. [11] For the Internet of Things, lightweight hybrid cryptography solutions, such as those combining Blowfish and AES, are recognized as effective for secure sensor data processing. [12]. In the Philippines, however, the adoption of hybrid Encryption remains limited; most studies concentrate on either AES or Blowfish individually rather than their combination. Nevertheless, compliance requirements under the Data Privacy Act of 2012 (RA 10173) and recent NPC Circulars[13][14] highlight the importance of implementing reasonable and appropriate encryption practices, making hybrid strategies increasingly relevant.

In Philippine institutions, Research Management Systems (RMS) are responsible for handling confidential materials like proposals, evaluations, and intellectual property documents. Presently, these systems frequently employ only one encryption method, most often AES-CBC or AES-ECB, without robust mechanisms for verifying data integrity. This can expose the systems to threats such as ciphertext alteration, replay attacks, and weaknesses stemming from improper key management. A study observed that solutions relying solely on AES or Blowfish, without the benefits of hybridization, are prone to performance issues when processing large datasets. [15]Similarly, research found that hybrid encryption schemes enhance performance in cloud service environments, implying that RMS could also benefit from these advanced frameworks. [16]

This research proposes a hybrid encryption approach named “CAGATCypher”, utilizing Blowfish for encrypting due to its speed. At the same time, AES-GCM wraps the Blowfish-encrypted data with its key diffusion to provide enhanced data integrity and confidentiality. This layered encryption strategy aligns with AEAD (Authenticated Encryption with Associated Data) standards and fulfills the requirements of RA 10173 for data security. Comparable hybrid frameworks have been successfully implemented in blockchain systems. [17] edge computing [18], and cross-platform healthcare RMSs [19]. By implementing this strategy, local institutional RMSs can benefit from improved efficiency and compliance with legal standards.

The relevance of this study stems from its capacity to advance both cybersecurity and regulatory adherence within Philippine academic institutions. For researchers, the proposed system enhances the confidentiality of document submissions, leveraging AES-GCM for tamper resistance and Blowfish for efficient processing. [5]Administrators and IT personnel benefit from improved scalability, as the hybrid system supports both legacy and modern infrastructures [8], [9], [15]. Compliance officers will find that the solution aligns with the provisions of RA 10173 and NPC Circulars, ensuring that RMS implementations meet the criteria for reasonable and appropriate protections. [13], [14]Governance authorities also benefit from having access to audit-ready, cryptographically secured data, which streamlines institutional reviews. [19], [20]Ultimately, integrating a hybrid encryption system places these institutions at the forefront of cryptographic standards, balancing security, efficiency, and compliance, consistent with global best practices.[11], [16], [17], [21].

In conclusion, this research tackles current shortcomings in Encryption by introducing a hybrid AES-Blowfish RMS framework designed to protect sensitive institutional and academic data. The system's multi-layered encryption and integrity verification features offer robust protection from data breaches, improve processing efficiency across varied workloads, and bring Philippine institutions up to both global and local data privacy standards. By addressing this gap, the study supplies a practical and forward-looking solution for stakeholders.

1.2 Literature Review

Relevance of Cybersecurity in Contemporary Times

Today, we rely more than ever on digital technology to make our lives and businesses run smoothly. While these advances bring faster processing and greater convenience, they also mean organizations are responsible for vast amounts of sensitive data. As a result, the risks of data breaches, identity theft, and hacking are always present [22][23]. That's why it's so important to have strong cybersecurity measures in place to address the challenges of our fast-changing digital world.

Even as digital security improves, cyber threats constantly evolve, making it challenging to stay ahead. Cybersecurity's primary goal is to keep data safe, even as risks change. But as technology advances, new problems pop up, and finding perfect solutions gets harder [24]. That's why we need clear laws and ethical guidelines—like data privacy acts—to help manage these new threats and protect everyone's information [13][14].

The fast-paced changes of the 21st century have made life more efficient, but they've also created new security threats. As a result, cybersecurity has become more critical than ever. With new technologies like AI, augmented reality, and the Internet of Things, organizations are always looking for ways to improve their operations. But along with these innovations comes the need to tackle new and more complex security issues [25].

IMPORTANCE OF ENCRYPTION

Encryption is all about taking information people can read (called plaintext) and turning it into something unreadable (called ciphertext) using math-based algorithms [26]. As our data needs have grown more complex, so have the tools to protect it. The Advanced Encryption Standard (AES) is one of the most trusted and efficient encryption methods, replacing older standards like DES. AES can use different key sizes—128, 192, or 256 bits—and its design makes it much safer than previous options. After rigorous testing, AES became the go-to standard for both classified and unclassified data. Researchers keep testing and improving it, even using things like neural networks and hybrid techniques [27]. In today's digital world, Encryption is a must for keeping sensitive data safe from prying eyes. There are two main types: symmetric Encryption (the same key for locking and unlocking data) and asymmetric Encryption (one public and one private key) [28]. Both types help ensure your data stays private, trustworthy, and only accessible by the right people. AES is fast and efficient, but sharing keys securely can be tricky. RSA, on the other hand, is great for secure key exchange but can be slower [29]. Studies show that hybrid approaches—using more than one method—might slow things down a bit, but they make your data much more secure.

IMPORTANCE OF DECRYPTION

Decryption is simply the process of turning encrypted data back into its original, readable form using a special key. Studies show that hybrid systems can strike a good balance between speed and security, even if they use a bit more memory. For example, depending on file size, hybrid Encryption might need between 21 and 74 MB of memory—but that extra effort pays off in much better data protection [30]. As hackers get smarter and computers get faster, older cryptography methods can become vulnerable. That’s why researchers are looking into new ideas like fuzzy logic and hypergraph theory to build even safer systems [31]. When choosing an encryption algorithm, it's essential to consider factors like speed, key size, and resistance to attacks to achieve the best mix of security and efficiency for your needs [32].

Decryption algorithms employ sophisticated cryptographic methods to protect the confidentiality of exchanged information. The combined use of Encryption and decryption is fundamental to secure communications. This overview examines AES's encryption and decryption processes, emphasizing their advantages, limitations, and key practical factors.

IMPORTANCE OF A HYBRID AES AND BLOWFISH

Mobile devices now serve a wide array of purposes beyond calls and messaging, including business transactions, banking, and academic functions through various applications. As a result, strengthening authentication security in mobile apps is crucial to prevent unauthorized access. A practical solution involves using a hybrid cryptographic approach that combines the Advanced Encryption Standard (AES) and Blowfish, both fast and powerful symmetric-key algorithms. Leveraging their large block sizes enhances the robustness of AES encryption, making attacks more difficult. [33]. In cloud computing, cryptography is essential for securing data in shared environments and addressing the complexities of migrating from traditional to cloud-based storage. To protect sensitive information, providers must implement advanced techniques that blend symmetric – and asymmetric Encryption [34]. For example, in law enforcement, sensitive data is encrypted, transferred, and decrypted only by authorized recipients, with hybrid cryptography offering an extra layer of protection during interdepartmental transfers. [35]. The expansion of cloud services and outsourced data has made safeguarding confidentiality, integrity, and security increasingly challenging. While RSA has long been used for these purposes, its need for large keys increases computational demands. Elliptic Curve Cryptography (ECC) has emerged as a preferred alternative, especially for resource-limited devices, providing strong security with fewer keys by exploiting the mathematics of elliptic curves. [36]. In some systems, documents are first encrypted with AES, and then the AES key is secured using Blowfish, ensuring the key is never exposed in plaintext and adding further protection.

Melina's system evaluation encompassed functionality, data security, and process efficiency through system function testing, measurement of encrypted data randomness (entropy test), file penetration using OpenSSL, and performance analysis in terms of processing time and file size. The research results show that the system successfully implemented double Encryption with a high entropy level (approaching 8.00) and resistance to penetration attacks.[37].

SIGNIFICANCE OF AES ALGORITHM

This application of the Advanced Encryption Standard demonstrates the AES algorithm's implementation for 128-bit data and a 128-bit key, using C software. To measure performance in the same system that uses the same hardware with a soft-core processor [38]. The presented designs are suitable for the encrypt-only AES-128 algorithm. Both designs integrate pipelining and iterative architectures into a single design. This is achieved through applying the concept of partial loop unrolling, where iterations and multistage pipelining are used to optimize area, throughput, and dynamic power consumption[39]. Nowadays, with the rapid development of multimedia technologies, research on safety and security is becoming more critical. Multimedia data are generated and transmitted through the communication channels and the wireless media [40]. In the realm of cybersecurity and information protection, symmetric cryptography stands as a foundation, securing the data and upholding the purity of confidentiality [41]. Encryption transforms original data into an unreadable form during transmission, while decryption restores it to its original state for the recipient. This guarantees that unauthorized parties cannot access the data. Cryptosystems have evolved, as communication technologies have rapidly advanced, stronger standards are needed. AES (Advanced Encryption Standard), based on the Rijndael algorithm, has become the current standard for encryption [42].

AES stands out for its reliability in consistently converting plaintext into ciphertext using a single key. It employs advanced encryption strategies, utilizing multiple processes and transformations to maintain high levels of security. In this context, an image encryption technique is introduced, using AES to address comprehensive security demands. This method ensures robust protection of digital images, making them resilient against attacks. The approach involves dividing the image into fragments before Encryption to enhance confidentiality further; upon decryption by the authorized recipient, these segments are reassembled to restore the original image. [43]

IMPORTANCE OF BLOWFISH ALGORITHM

Cryptographic operations often require substantial computing resources. Blowfish, however, offers an efficient solution for encrypting data across different settings, including IP-based Internet of Things (IoT) networks. [44]As electronic data exchange increases and multimedia applications proliferate, ensuring the security of transmitted and stored information, particularly images, has become a top priority. [45]

Cloud computing refers to interconnected computing resources accessed over the internet, offering cost-effective processing and storage. Its rapid growth has led to significant interest and research. Protecting personal information in the cloud is challenging due to the need for confidentiality, authenticity, accessibility, data integrity, and recovery. Hybrid encryption approaches, as discussed in this study, address a wide range of security challenges within cloud environments. [46] Blowfish, a symmetric key algorithm, uses the same secret key for both encrypting and decrypting data, unlike DES, which utilizes a Feistel structure with separate subkeys. Performance assessments show that Blowfish achieves faster Encryption and decryption times (0.2ns-1.2ns) compared to DES (2ns-3ns) and consumes less memory, making it a resource-efficient choice. [47]

IMPORTANCE OF A HYBRID ENCRYPTION ALGORITHM

Hybrid encryption algorithms are designed to overcome the limitations of relying on a single encryption method and to enhance the security of conventional systems. By analyzing established algorithms, refining their features, and arranging them in specific sequences, these hybrids significantly bolster data protection. For example, some systems combine original algorithms like Micro Genard and Base64 to produce more secure hybrid Encryption. [48]. A systematic review of hybrid encryption solutions across various platforms, including AES and Blowfish combinations, reveals notable improvements in both security and performance. [49]. Research highlights the use of AES-Blowfish hybrids for mobile app authentication, exploiting their large block sizes to defend against man-in-the-middle attacks. Enhanced models sometimes implement modified AES S-boxes with Blowfish and integrate ECC for secure key exchange and digital signatures, optimizing results for multiple file types. The proposal presents an enhanced security model using modified AES with altered S-boxes, combined with Blowfish. It also includes ECC for key exchange and digital signatures, evaluating performance across different file formats.[50] It addresses cloud data security by implementing a hybrid Dynamic AES (DAES) and Blowfish structure to protect against potential brute force and algebraic attacks while maintaining processing speed.[51] Other projects focus on low-power, high-throughput AES-Blowfish hybrids to ensure safe data transmission, particularly in internet and mobile storage scenarios.

COMPARISON OF HYBRID AES & CAGATCipher

Traditional encryption systems, such as DES and Triple DES, face challenges when applied to images due to their high redundancy and large size. To address these challenges, this paper explores secure image encryption by comparing two hybrid cryptosystems, AES-256-GCM-SHA384 and ChaCha20-Poly1305-SHA256. Implemented locally in Python. This work evaluates the performance and security of these systems using metrics including Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), Structural Similar-ity Index (SSIM), entropy, total encryption/decryption process time, Number of Pixel Change Rate (NPCR), Unified Average Changing Intensity (UACI), correlation coefficients, and Bit Error Rate (BER).[5] It handles data security concerns in shared environments and the difficulties that arise when businesses move from traditional storage to cloud storage, cryptography is essential for secure cloud computing. Therefore, to safeguard sensitive data, cloud service providers must employ advanced methods like symmetric and asymmetric key cryptography. This article examines a number of cryptographic systems, pointing out the shortcomings of existing methods and emphasizing the superior efficiency of hybrid cryptography, which mixes symmetric and asymmetric encryption. Researchers frequently pair AES with Rivest-Shamir-Adleman and Elliptic Curve Cryptography algorithms, emphasizing that alternatives like Twofish and Blowfish provide reliable security and effectiveness. In this study, a 128-bit key was used to evaluate the performance of Blowfish, Twofish, and AES. [6] Lightweight Cryptography Algorithms (LWC), which aim to maximize data security while minimizing resource consumption during the encryption and decryption operations, are among the algorithms that try to prevent unauthorized people from decoding encrypted data. Integration is made easier by connecting to a wide range of Internet of Things (IoT) devices and communication protocols. This study recommends enhancing the security of the IoT platform by utilizing a hybrid cryptography technique.

Two separate algorithms, AES (SPN-based structure) and Camellia (FN-based structure), are combined in the proposed technique. The NIST-800-22 test suite demonstrates how effectively it functions in a variety of scenarios, including observed improvement in six different statistical tests, and from this study, we can conclude that the hybrid algorithm outperforms its parts individually in terms of performance. Consequently, it may be inferred that the hybrid algorithm offers enhanced security.[7]

COMPLEMENTING KEY MANAGEMENT

Complementary key management has been identified in cryptographic literature as a vital enhancement to symmetric encryption systems, emphasizing that overall security depends not only on the cipher but also on proper key handling. NIST recommends the use of structured key derivation and key separation mechanisms to limit key reuse and reduce the impact of key compromise, as outlined in SP 800-56C and related publications. An AES-based Key Derivation Function (KDF) leverages the strong diffusion and one-way properties of AES to securely derive session and sub-keys from a master key, providing efficient and cryptographically sound key separation. Ferguson, Schneier, and Kohno further note that hierarchical key management and frequent key rotation are effective defenses against practical attacks arising from long-term key exposure and implementation flaws, even when robust encryption algorithms are employed [61]–[63].

1.3 Objective of the Study

General Objective

To develop a new hybrid encryption algorithm that combines the AES and Blowfish encryption algorithms, creating an algorithm that is aiming to be more secure (modified encryption & key fuzzing) and faster (optimize the algorithm overall) from other typical algorithms.

Specifically, it aims:

1. To effectively determine the optimal performance after implementing the hybrid AES and Blowfish encryption, using speed and responsiveness as key metrics.
2. To compare the overall performance of the CAGATCipher compared to other hybrid and standard encryption algorithm.
3. To implement effective key management to ensure efficient and secure handling of Encryption within institutional workflows.

1.4 Scope and Limitations

This study aims to assist the research office at Saint Michael College of Caraga in designing, developing, and implementing a hybrid encryption algorithm that combines the Blowfish and Advanced Encryption Standard (AES) algorithms to enhance the security of institutional systems. This system is designed for the faculty and students of SMCC and, therefore, specifically caters to their data confidentiality. The primary objective is to assess how well the proposed hybrid approach protects sensitive institutional data during transmission and storage. The system is evaluated in a research office typical of administrative or educational organizations, considering factors such as data confidentiality, encryption/decryption speed, and resilience to common cryptographic threats.

Although the Hybrid Encryption study is confined to the examination of AES and Blowfish algorithms exclusively, it has limitations. The system is for the research office only and not for the whole institution. However, other symmetric or asymmetric encryption techniques are tested in a controlled setting, so they might not work as well in real-life or extreme threat situations. The study also does not examine management strategies or how to connect to large institutional networks, which may require additional structure and security measures. Lastly, the system is explicitly designed for Saint Michael College of Caraga, which restricts its applicability to other institutions or regions.

1.5 Definition of Terms

The terms defined below explain how we used this information in our study and describe the fundamental principles and factors considered throughout the development and comparison of the Hybrid Encryption Algorithm (BlowFish & AES)**.**

Advanced Encryption Standard (AES) – It is an encryption algorithm that is widely used worldwide. Formerly known as the Rijndael cipher, it was chosen as the Encryption Standard during the Advanced Encryption Standard (AES) Competition. It is one of the two main algorithms we will utilize, and it is vital for developing more optimized encryption.

Algorithm – The core fundamentals of the research. An algorithm forms the entire system for it to be operational, and we are trying to develop the best possible method for such Encryption. It is vital for the development.

Blowfish Algorithm – Another main encryption algorithm that we are going to utilize. Although it is not chosen as the standard encryption algorithm, it still has its advantages compared to AES. We aim to combine the efficiency of the two to develop a faster and more efficient hybrid encryption algorithm.

Decryption – The retrieval of the encrypted data. It is how we recover the encrypted data from the Database to its original state. Decryption happens when the authorized user requests data, and the algorithm decrypts it from the Database and presents it to the user in the original state.

Encryption – The essence of the research. It is how we manipulate the data to protect it from unauthorized threats. It should be unrecognizable by regular users and (near) impossible to decipher. Depending on how we handle this process, we output a certain encryption based on the preferred Encryption Algorithm.

Execution and Time – It is how long a specific process in a code has been running before ending. This data is necessary to measure how fast an algorithm processes the data based on its data type and size. The two algorithms (Blowfish & AES) will be measured this way, and if recognizable and precise patterns emerge, the favored algorithm will be used in that state.

AES-Based Key Derivation Function - uses the AES block cipher as a deterministic one-way transformation to generate new cryptographic keys from a master key and contextual inputs such as nonces, counters, or identifiers. By encrypting structured input data with AES, the derived keys inherit AES's strong diffusion and resistance to cryptanalysis, ensuring key separation and preventing key reuse. This approach enables secure session, sub-key, or per-block key generation with minimal performance overhead.

XOR – This is the method used for the Encryption. It is mainly used in both AES and Blowfish and is vital for their complex encryptions. XOR is the manipulation of data at the binary level, meaning the binary representation of the data is used for Encryption. It is also one of the logic gates in computing.

CHAPTER II

METHODOLOGY

2.1 Research Design

Developmental-Experimental Design

This study will adopt a developmental-experimental research design, which combines the design and implementation of a technical solution (AES & Blowfish integration) with testing and evaluation under real or simulated conditions. The developmental aspect focuses on designing the encryption framework, while the experimental aspect measures its effectiveness and performance.

The study will utilize a quantitative method to collect and analyze numerical data regarding:

- System performance metrics (e.g., CPU usage, response time, memory load)
- Encryption/decryption time
- Data access success/failure rates before and after Encryption
- User-level access and data breach simulation results

These quantitative results will support a measurable evaluation of the AES & Blowfish implementation's impact.

This research follows an applied approach, aiming to solve a practical, real-world problem—securing institutional systems from data breaches and unauthorized access—by implementing an encryption-based solution. The focus is on practical implementation, testing, and usability, rather than purely theoretical exploration.

The research uses the following principles of the algorithms: Advanced Encryption Standard and Blowfish.

- **Advanced Encryption Standard process:**

Step 1: Pad the plaintext so its length is a multiple of 16 bytes (block size).

Step 2: Split each 16-byte block into a 4×4 state matrix (fill column-wise).

Step 3: Choose the key (128/192/256 bits) and run Key Expansion to produce all round keys.

Step 4: XOR the state with RoundKey 0 (AddRoundKey — the initial whitening).

Step 5: For each round i = 1 to Nr−1 (Nr = 10/12/14 for 128/192/256-bit keys) do:

- 5.1: SubBytes — substitute every byte using the S-box.
- 5.2: ShiftRows — cyclically rotate each row (row r by r bytes left).
- 5.3: MixColumns — apply the linear GF(2^8) column transformation.
- 5.4: AddRoundKey — XOR the state with RoundKey i.

Step 6: Final round (round Nr) — do SubBytes, then ShiftRows, then AddRoundKey (skip MixColumns).

Step 7: Read out the state matrix column-wise → that 16-byte result is the ciphertext block.

Step 8: If multiple blocks (and using a block-cipher mode), process remaining blocks according to the chosen mode (ECB/CBC/CTR/etc.).

- **Blowfish process:**

Step 1: Provide inputs

- Supply a user key (4–56 bytes).
- Supply plaintext to encrypt.

Step 2: **Convert inputs to bytes**

- Convert key string → key bytes.
- For encryption, split plaintext into 64-bit (8-byte) blocks; pad the last block (e.g., PKCS#5/7 or zero padding).

Step 3: **Load initial constants**

- Initialize the P-array (18 × 32-bit words) and 4 S-boxes (4 × 256 × 32-bit) with fixed constants (hex digits of π).

Step 4: **XOR the key into the P-array**

- Cycle through key bytes 4 at a time, form 32-bit words and XOR them with P[0]..P[17] (wrap key bytes if shorter).

Step 5: **Prepare an all-zero 64-bit block**

- Set Left = 0x00000000, Right = 0x00000000.

Step 6: **Iteratively encrypt to replace P-entries**

- Encrypt the current 64-bit block using the current P and S.
- Replace P[0] and P[1] with the encryption output (L_out,R_out).
- Repeat (encrypt zero again) to replace P[2],P[3], … until all 18 P entries are replaced. *(This requires 9 encryptions for P-array.)*

Step 7: **Iteratively encrypt to fill S-boxes**

- Continue encrypting the most recently produced 64-bit output and write results pairwise into S1[0],S1[1], S1[2],S1[3], … through all 4 S-boxes. *(This requires 512 encryptions for S-boxes.)*

Step 8: **Finish key schedule**

- After total **521** encryptions, the P-array and S-boxes are fully key-dependent — key schedule complete.

Step 9: **Prepare a single plaintext block for encryption**

- Take an 8-byte (64-bit) plaintext block. Split into Left (L) and Right (R), each 32 bits.

Step 10: **Define the F-function**

F(x):

a = (x >> 24) & 0xFF

b = (x >> 16) & 0xFF

c = (x >> 8) & 0xFF

d = x & 0xFF

return ((S1[a] + S2[b]) XOR S3[c]) + S4[d] (mod 2^32)

Step 11: **Perform 16 Feistel rounds (encryption)**

For i = 0 to 15 (rounds 1..16):

- L = L XOR P[i]
- R = R XOR F(L)
- Swap(L, R)

Step 12: **Finalize the block after rounds**

- Undo the final swap (swap back).
- R = R XOR P[16]
- L = L XOR P[17]
- Ciphertext block = concatenate(L, R)

Step 13: **Process all blocks**

- Repeat Steps 9–12 for every 8-byte block (with chosen chaining mode if encrypting >1 block).

Step 14: **Decryption**

- Same operations but P-array is used in reverse order: use P[17],P[16], … P[0] in the round XORs (effectively reversing the per-round P values). F function and swap behavior are identical.

Step 15: **Notes & caveats**

- **Block size:** 64 bits — beware birthday limit for very large volumes of data.
- **Key schedule cost:** extraordinarily slow (521 full encryptions). Do key schedule once per key and reuse subkeys.
- **Security:** S-boxes and P-array are fully replaced by key schedule; small changes in key produce large differences.
- **Padding/Mode:** Choose padding (PKCS#5/7) and an appropriate mode (CBC, CTR, GCM via other ciphers) for multi-block messages.

- **CAGATCipher process**:

(Initial AES Process)

Step 1: Pad the plaintext so its length is a multiple of 16 bytes (block size).

Step 2: Split each 16-byte block into a 4×4 state matrix (fill column-wise).

Step 3: Get the key for AES from the first 128-bits of the key and run Key Expansion to produce all round keys.

Step 4: XOR the state with RoundKey 0 (AddRoundKey — the initial whitening).

Step 5: For each round i = 1 to Nr−1 (Nr = 10/12/14 for 128/192/256-bit keys) do:

• 5.1: SubBytes — substitute every byte using the S-box.

• 5.2: ShiftRows — cyclically rotate each row (row r by r bytes left).

•5.3: LiteMixColumns — apply the modified lite version of GF(2^8) column transformation.

• 5.4: AddRoundKey — XOR the state with RoundKey i.

Step 6: Final round (round Nr) — do SubBytes, then ShiftRows, then AddRoundKey (skip MixColumns).

Step 7: Read out the state matrix column-wise → that 16-byte result is the ciphertext block.

Step 8: If multiple blocks (and using a block-cipher mode), process remaining blocks according to the chosen mode (ECB/CBC/CTR/etc.).

(Blowfish Process)

Step 9: **Provide inputs**

- Supply a user key (4–56 bytes or 32-448 bits of data after the first 128 bits of the key).
- Supply encrypted text from the AES process to encrypt.

Step 10: **Convert inputs to bytes**

- Convert key string → key bytes.
- For encryption, split plaintext into 64-bit (8-byte) blocks; pad the last block (e.g., PKCS#5/7 or zero padding).

Step 11: **Load initial constants**

- Initialize the P-array (18 × 32-bit words) and 4 S-boxes (4 × 256 × 32-bit) with fixed constants (hex digits of π).

Step 12: **XOR the key into the P-array**

- Cycle through key bytes 4 at a time, form 32-bit words and XOR them with P[0]..P[17] (wrap key bytes if shorter).

Step 13: **Prepare an all-zero 64-bit block**

- Set Left = 0x00000000, Right = 0x00000000.

Step 14: **Iteratively encrypt to replace P-entries**

- Encrypt the current 64-bit block using the current P and S.
- Replace P[0] and P[1] with the encryption output (L_out,R_out).
- Repeat (encrypt zero again) to replace P[2],P[3], … until all 18 P entries are replaced. *(This requires 9 encryptions for P-array.)*

Step 15: **Iteratively encrypt to fill S-boxes**

- Continue encrypting the most recently produced 64-bit output and write results pairwise into S1[0],S1[1], S1[2],S1[3], … through all 4 S-boxes. *(This requires 512 encryptions for S-boxes.)*

Step 16: **Finish key schedule**

- After total **521** encryptions, the P-array and S-boxes are fully key-dependent — key schedule complete.

Step 17: **Prepare a single plaintext block for encryption**

- Take an 8-byte (64-bit) plaintext block. Split into Left (L) and Right (R), each 32 bits.

Step 18: **Define the F-function**

F(x):

a = (x >> 24) & 0xFF

b = (x >> 16) & 0xFF

c = (x >> 8) & 0xFF

d = x & 0xFF

return ((S1[a] + S2[b]) XOR S3[c]) + S4[d] (mod 2^32)

Step 19: **Perform 16 Feistel rounds (encryption)**

For i = 0 to 15 (rounds 1..16):

- L = L XOR P[i]
- R = R XOR F(L)
- Swap(L, R)

Step 20: **Finalize the block after rounds**

- Undo the final swap (swap back).
- R = R XOR P[16]
- L = L XOR P[17]
- Ciphertext block = concatenate(L, R)

Step 21: **Process all blocks**

- Repeat Steps 9–12 for every 8-byte block (with chosen chaining mode if encrypting >1 block).

Step 22: **Decryption**

- Same operations but P-array is used in reverse order: use P[17],P[16], … P[0] in the round XORs (effectively reversing the per-round P values). F function and swap behavior are identical.

2.2 System Architecture

This chapter analytically examines the technical requirements in the design and development of “A Hybrid Encryption Algorithm Using Advanced Encryption Algorithm and Blowfish for Institutional System”. Aspects include the software and hardware requirements needed for the system's proper functioning. This chapter also explains the system's features. It focuses, as a whole, on applying the Encryption method algorithm in evaluating the research documents for staff and students of Saint Michael College of Caraga.

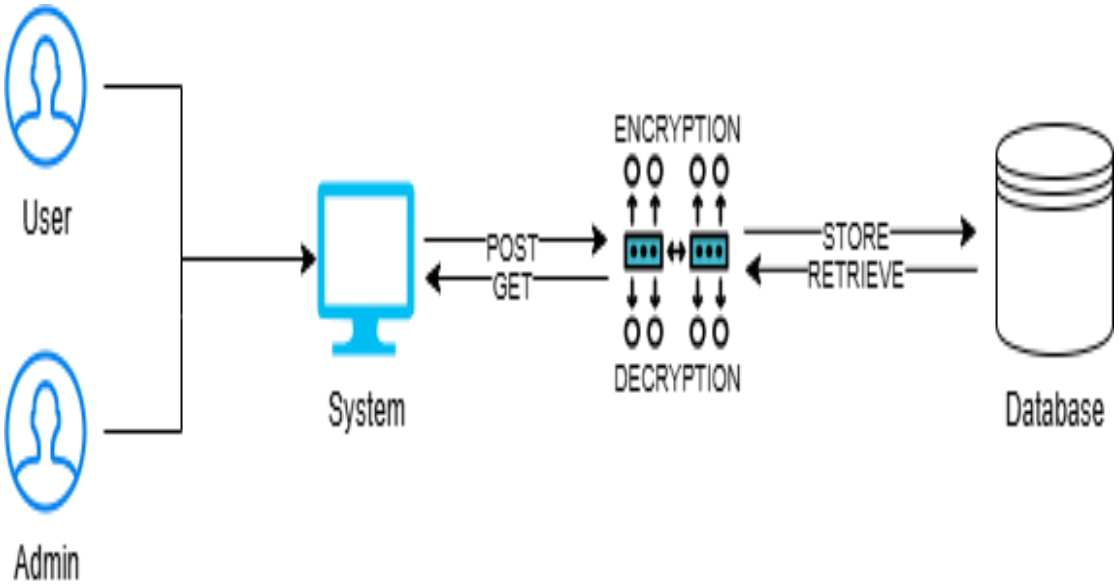


Figure 1. System Architecture

Figure 1 explains how the Admin and user interact with the system. The Admin manages information such as encrypted documents or updates, which the system processes and stores in the Database. The Admin oversees and reviews this information through the Admin Dashboard. The users can only access their respective accounts and have full access to them.

2.3 Activity Diagram

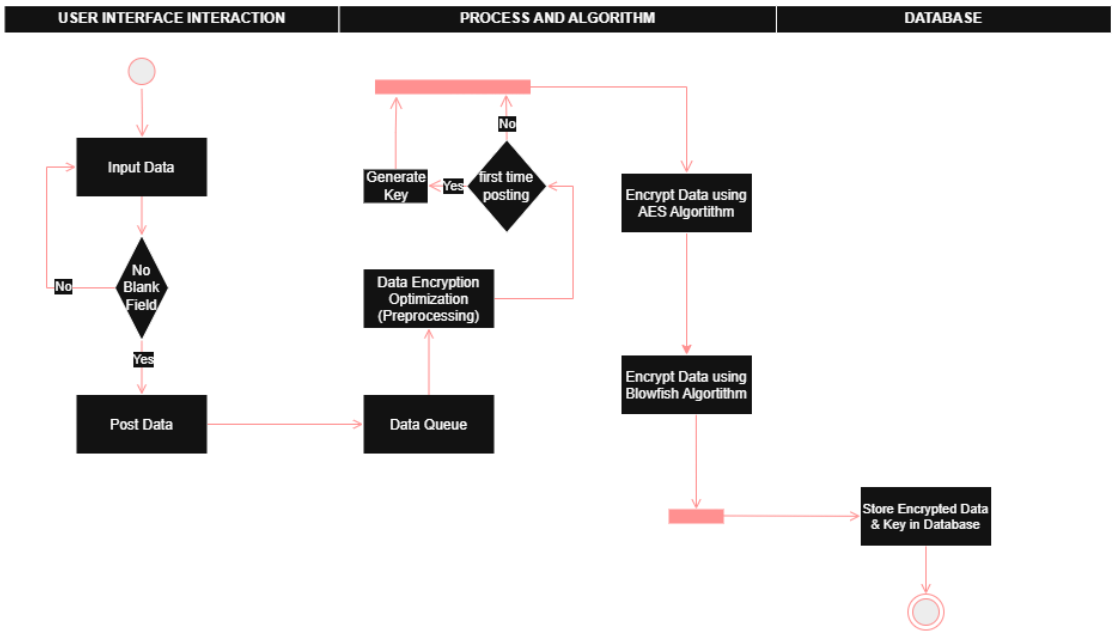


Figure 2: Activity Diagram – *Post Data*

Figure #2 explains the series of actions taken by the user (student, adviser, panel, dean & Principal, and Admin) to the system. The user accesses the website and inputs the desired information or file. It'll then be processed by the algorithm after submitting/posting. The algorithm will first preprocess the data in preparation for decryption. It'll then generate the corresponding keys for the encryption process. After that, it will encrypt the data using the two algorithms and store it in the Database.

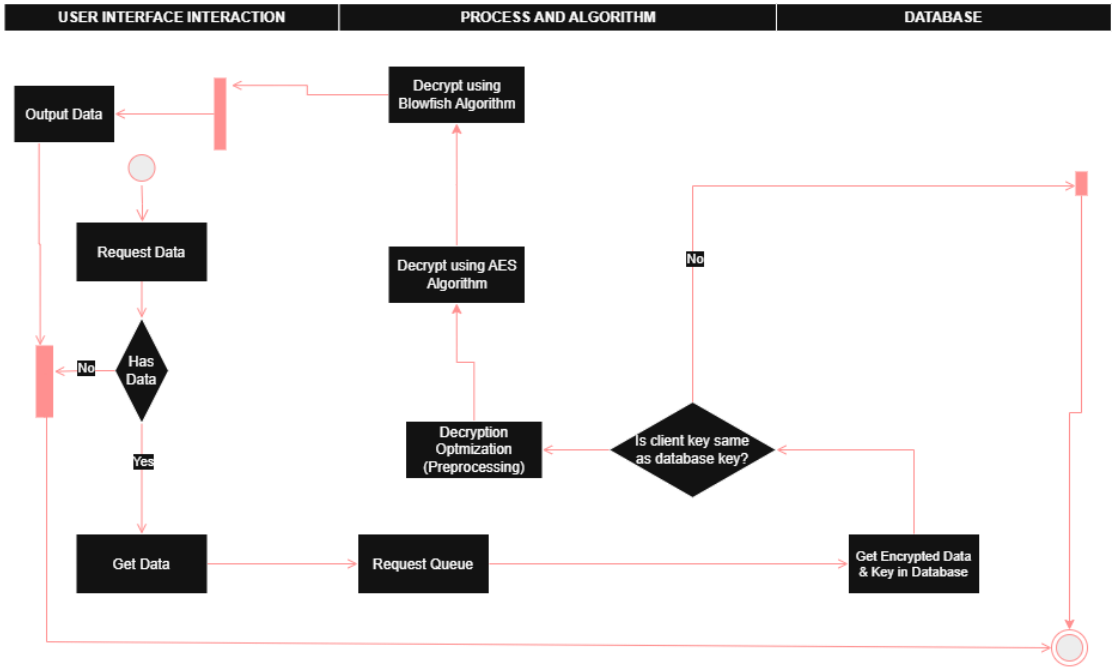


Figure 3: Activity Diagram – *Get Data*

Figure #3 presents the series of activities taken by the user (student, adviser, panel, dean & Principal, and Admin) to the system. It's similar to the previous Diagram, but instead of posting data, we request the data. After the user requests data, the process goes to the queue to avoid overloading the system. After its turn, the system will get the key from the Database and then compare it to the client's key. If it doesn't match, it'll throw out of the process. Otherwise, the system will start the decryption process by preprocessing the encrypted data. It will then use the two series of algorithms to decrypt said data. It would then be output depending on the system's User Interface, be it visually or in a file.

2.4 Conceptual Framework

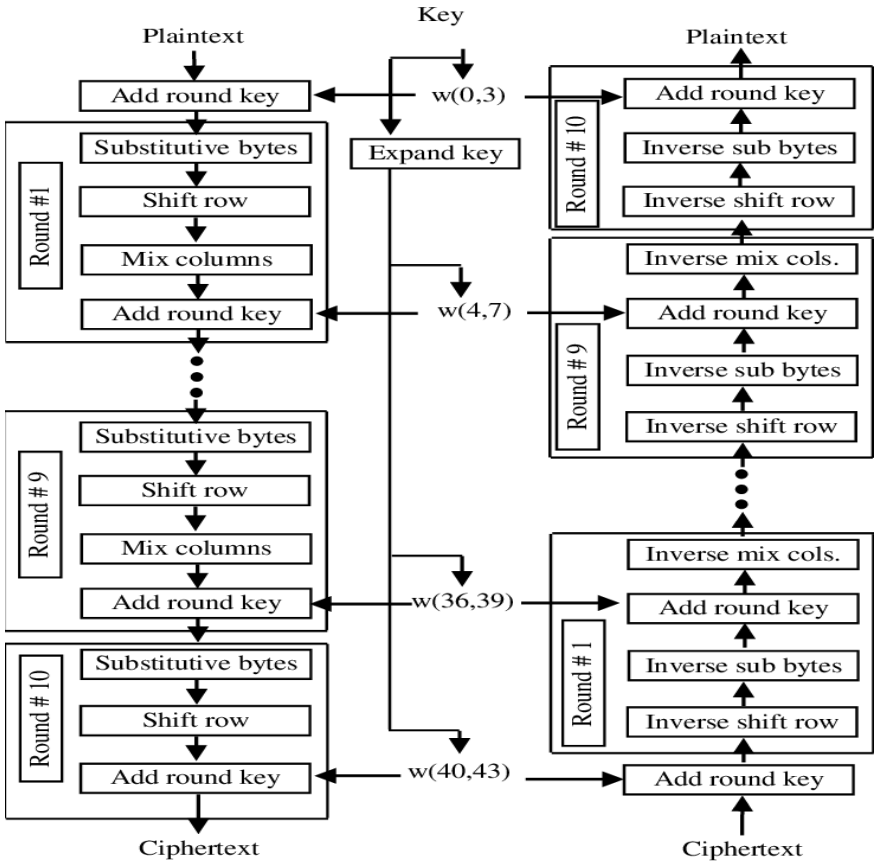


Figure 4: Advanced Encryption Standard Diagram

Figure # shows the process of AES encryption algorithm. It has robust process and rounds that will ensure that the data would be encrypted beyond recognition by any cracking software. The algorithm is repeated through rounds and each rounds has these following processes: Substitution bytes, Shift row, Mix columns, and Add round key. Also note that this entire process would be reversed when decryption happens. After some tests and research, we also found that the “Mix columns” part of every round takes a significant amount of processing time.

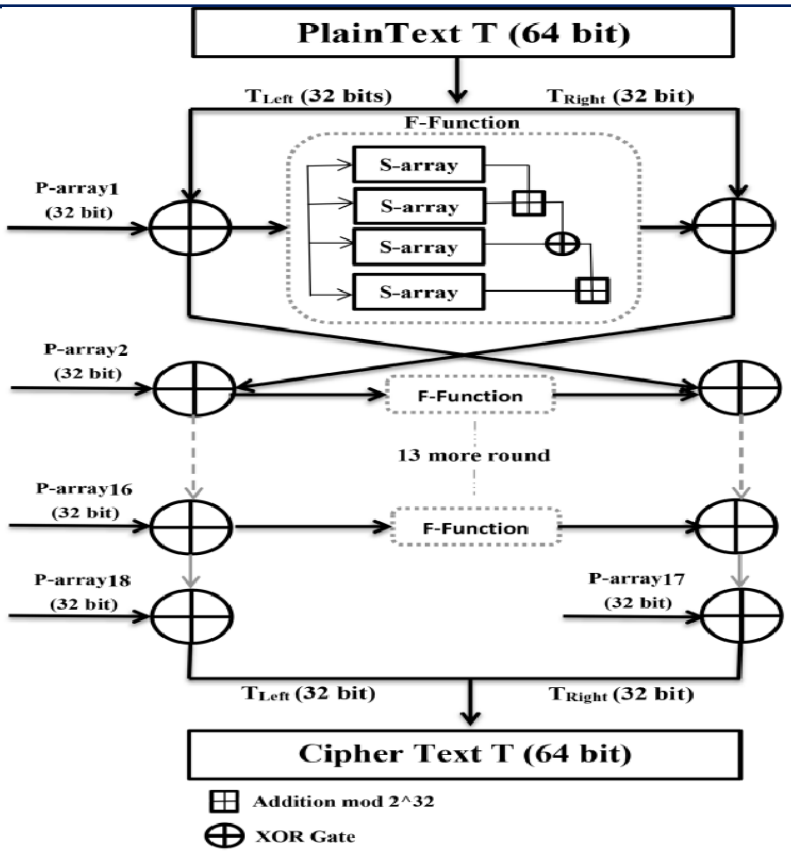


Figure 5: Blowfish Diagram

Figure # shows the process of Blowfish encryption algorithm. It has a simple and straightforward encryption process that is repeated 16 times to come up with the encrypted text with the use of a P-array and S-array. Also note that every rounds, the key would be reprocessed unlike AES. Though from the entire process, Blowfish is significantly faster than AES with algorithm runtime. Blowfish also features a varying key size ranging from 32 bits to 448 bits unlike AES which only has 128-bits, 192-bits, and 256-bits version.

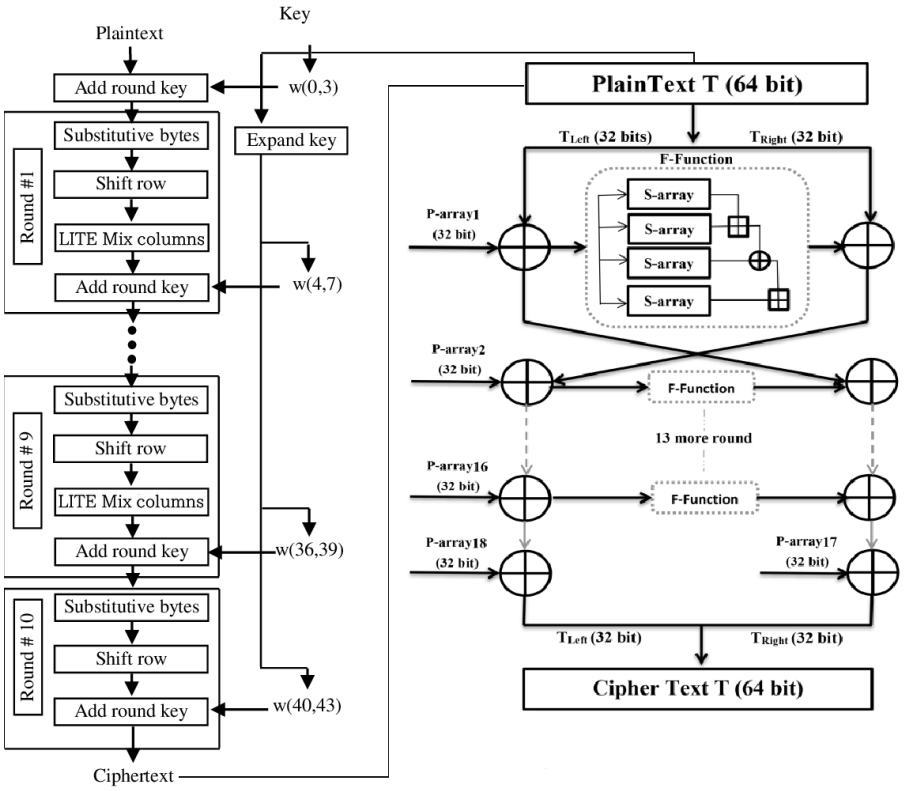


Figure 6: CAGATCipher Conceptual Framework

Figure # shows our proposed algorithm that combines the use of AES and Blowfish encryption. This combination of algorithm can help secure the data further at the same time having a faster encryption than the standard AES. The keys that can be used in this algorithm range from 160-bits to 576-bits. The key is used in sections by the two algorithms. The first 128-bits of the key would be used by the AES, the rest of the key is to be used by the Blowfish algorithm (32-bits to 448-bits). Furthermore, to make the combined algorithm run much faster, we reduced the process of the AES, specifically its process demanding instruction in the "MixColumns" section. Instead of using the AES's standard GF(2^8) finite field multiplication, we will modify and reduce its multiplication process while not significantly affecting the diffusion of the data. The modification made is shown below:

$$\begin{bmatrix} 02\ 03\ 01\ 01 \\ 01\ 02\ 03\ 01 \\ 01\ 01\ 02\ 03 \\ 03\ 01\ 01\ 02 \end{bmatrix} 2 \cdot a = \begin{cases} 1 \cdot a = a \\ (a \ll 1)\ if\ a < 0x80 \\ (a \ll 1) \oplus\ 0x1B\ if\ a \geq 0x80 \\ 3 \cdot a = (2 \cdot a) \oplus a \end{cases}$$

b0=(02·a0)⊕(03·a1)⊕(01·a2)⊕(01·a3)

b1=(01·a0)⊕(02·a1)⊕(03·a2)⊕(01·a3)

b2=(01·a0)⊕(01·a1)⊕(02·a2)⊕(03·a3)

b3=(03·a0)⊕(01·a1)⊕(01·a2)⊕(02·a3)

Figure 7: GF(2^8) matrix and its multiplication rules *(above)* and GF(2^8) formula/conversion *(below)*

The standard AES Matrix used for "MixColumns" is a bit demanding in terms of resources and time. So we modified it so that it'll be lighter at the same time have enough diffusion capabilities for it to be acceptable (considering we would use Blowfish on top of it). The modified version is shown below:

$$\begin{bmatrix} 02\ 03 \\ 01\ 02 \\ 03\ 01 \\ 02\ 03 \end{bmatrix} 2 \cdot a = \begin{cases} 1 \cdot a = a \\ (a \ll 1)\ if\ a < 0x80 \\ (a \ll 1) \oplus\ 0x1B\ if\ a \geq 0x80 \\ 3 \cdot a = (2 \cdot a) \oplus a \end{cases}$$

b0=(02·a0)⊕(03·a1)

b1=(01·a1)⊕(02·a2)

b2=(03·a2)⊕(01·a3)

b3=(02·a3)⊕(03·a0)

Figure 8: Lite version of GF(2^8) matrix and its multiplication rules *(above)* and its formula/conversion *(below)*

2.5 Data Gathering Procedure

The data we will test and experiment with is primarily the information within the institution's Research Management System. It comprises datasets that we can analyze and manipulate. The performance and results would be consolidated for the final interpretation and development of the optimal implementation. The data should be of varied types and sizes to achieve diverse results, providing more flexibility and integrity to the application of algorithms.

2.6 Algorithms Used

Encryption has been widely utilized in every system, becoming standard for data protection. The arrays of security measures developed over the years result from the continuous iteration of exploiting vulnerabilities for harmful gain. With that, we need to improve specific measures to remain relevant to security trends and current standards. Encryption is our target as it is widely used for system security and serves as the core of what we are trying to protect: the data. We have two main algorithms that we can try to utilize and improve, mainly the AES and Blowfish encryption algorithms.

Advanced Encryption Standard (AES) Algorithm

The Advanced Encryption Standard is the most utilized encryption algorithm because it is not only efficient and sustainable but also simplistic. [59] From the encryption standard competition by the US National Institute of Standards and Technology (NIST), the Rijndael algorithm emerged from a series of tests prepared and conducted by the NIST. [60]. Over the years since its development, the integrity of the algorithm has remained prevalent and widely used in existing systems. This algorithm can cover most types of data as it is recent and supports a variety of modern hardware innovations.

Blowfish Algorithm

In some cases, businesses still use old and legacy hardware/software. Encrypting such data in that old software is overkill with modern algorithms like AES. The Blowfish algorithm, which is made for older software, is much more efficient and uses less memory compared to AES. The Blowfish algorithm is old but designed for legacy 32-bit systems, which are more compatible and better suited to this hardware. The newer version, Twofish, expands its block cipher to 128 bits but is slightly slower. Depending on the efficiency of the algorithm, we can utilize them alternatively for better performance.

2.7 Use Case Diagram

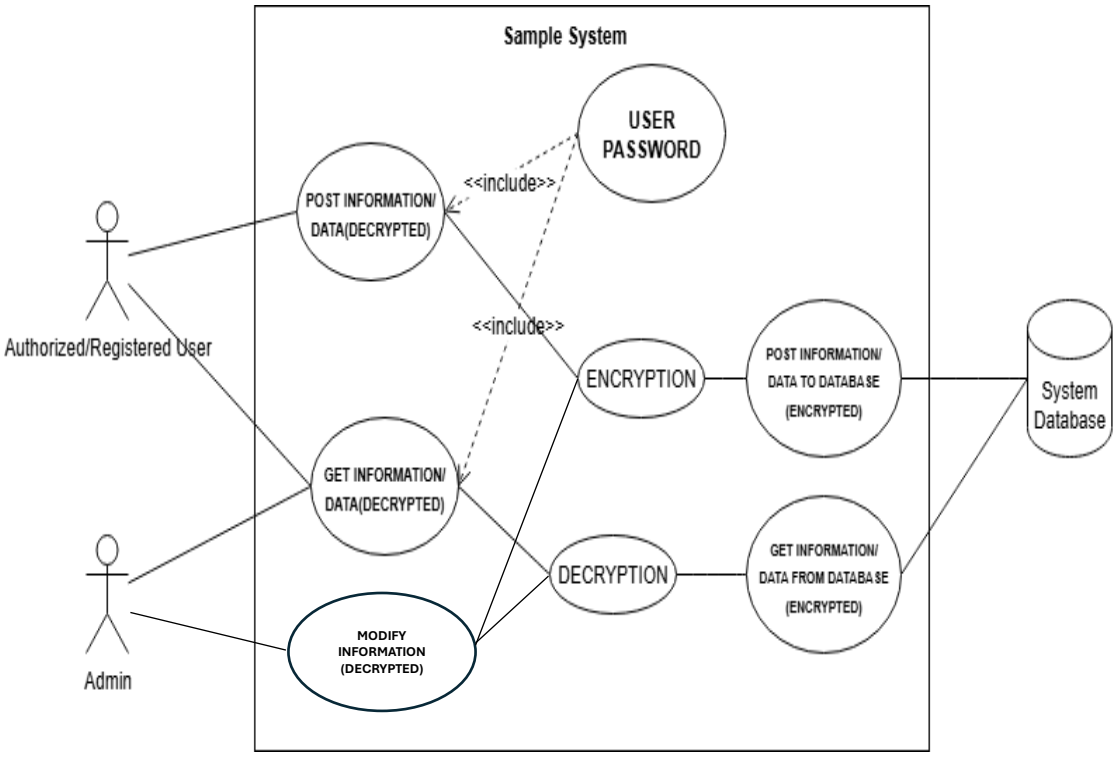


Figure 9: Use Case Diagram

Figure 6 illustrates who can access each function in the system and stresses the interactions between the system and its users. The user can post or get data from their account, and the system can encrypt or decrypt the data, respectively. Posting and retrieving information can only be done if the user has sufficient credentials (e.g., password) to access the account's database (Students, Adviser, Dean & Principal, and Admin). It also incorporates an admin system where the Admin can directly access and modify user information. The entire algorithm depends on the type of architecture/system being used, whether it be a legacy system or a modern one.

2.8 Sequence Diagram

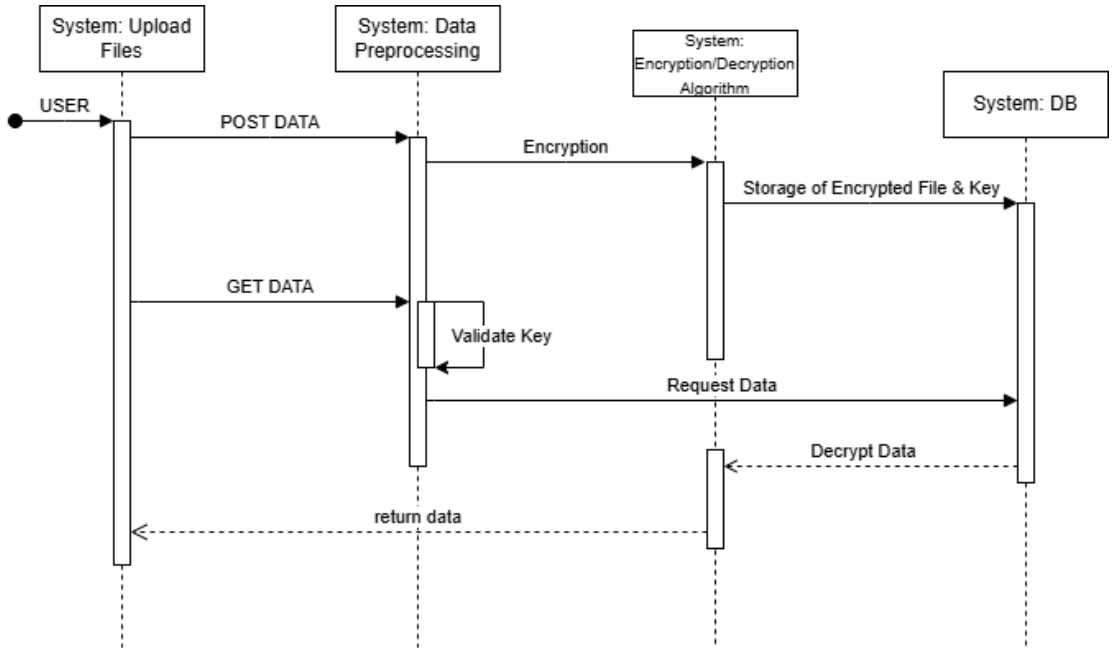


Figure 10: Sequence Diagram

In Figure 7, our Sequence Diagram is straightforward. It applies to every account (Students, Adviser, Dean & Principal, and Admin). The user posts data to the system through its UI, and the system will then start by preprocessing the data. From that, it'll encrypt the data along with the key, use a series of algorithms, and then store it in the Database. In contrast, requesting data requires the user or the client server to have the appropriate key to pull the data from the Database.

2.9 ERD (Database Design)

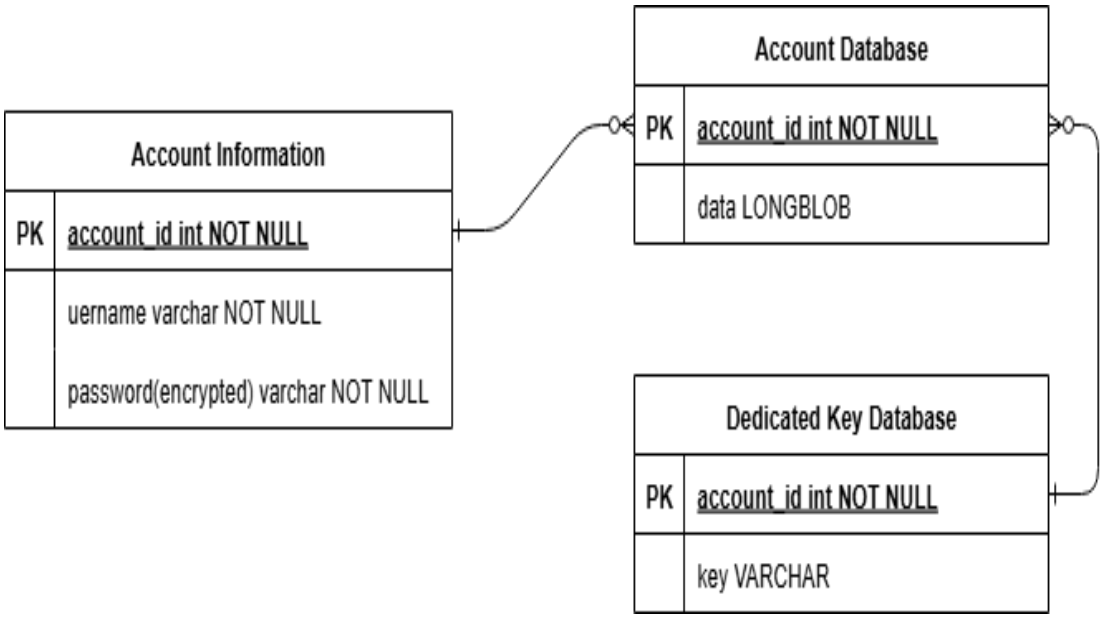


Figure 11: Entity Relationship Diagram

In Figure 8, the algorithm will use this Entity Relationship Diagram with three central databases for the respective accounts (Student, Adviser, Dean & Principal, Panel, and Admin): the Account Information, Account Database, and Dedicated Key Database. Account Information stores the account credentials, used to log in to the account. The password is encrypted o protect against breaches or leaks. The Account Database stores all the data used by that account. This can range from simple text and values to a whole file or documents. The Dedicated Key Database is where the keys to a corresponding account are stored. The keys are to be pulled if requests for decryption are called.

2.10 User Interface Design (Prototype)

The research focuses only on the algorithm. The user interaction is entirely dependent on the system it was integrated into, so an integrated UI isn't necessary. Our target system for testing is the institution's Research Management System, and the algorithm operates in the background. However, to simulate the flow of data and the Database, we have to create a prototype that mimics the Research Management System of the Institution.

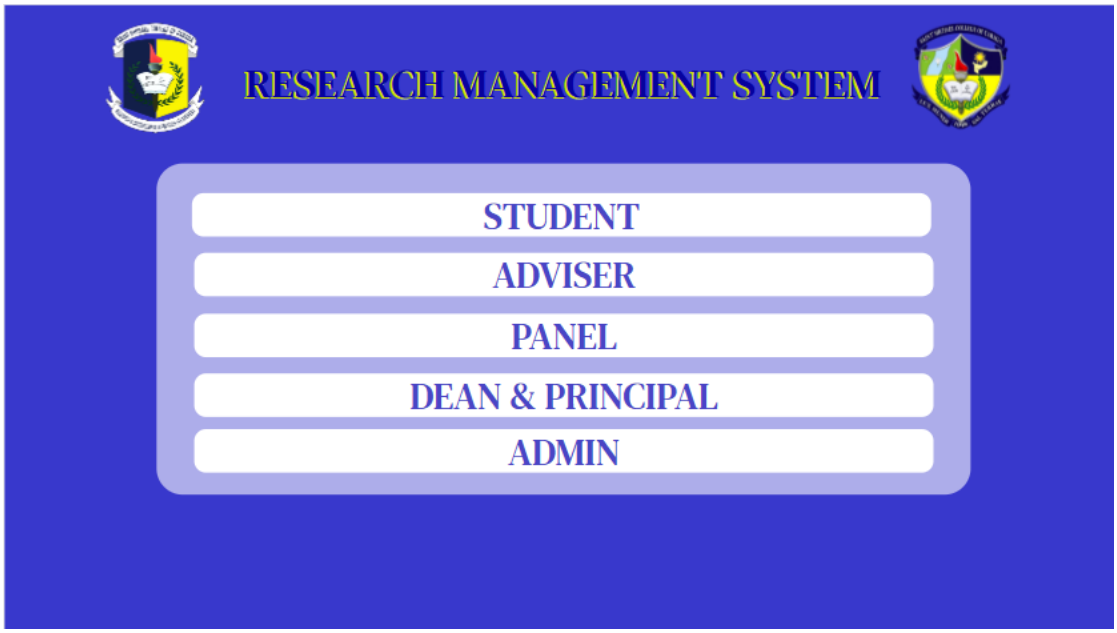


Figure 12: Login Dashboard

The Figure above shows the login page for the simulated system. The User Interface will be similar to the Research Management System used in the institution to produce accurate results. The login landing page consists of the Student, Adviser, Panel, Dean & Principal, and Admin buttons, which will direct you to the respective login pages for each account.

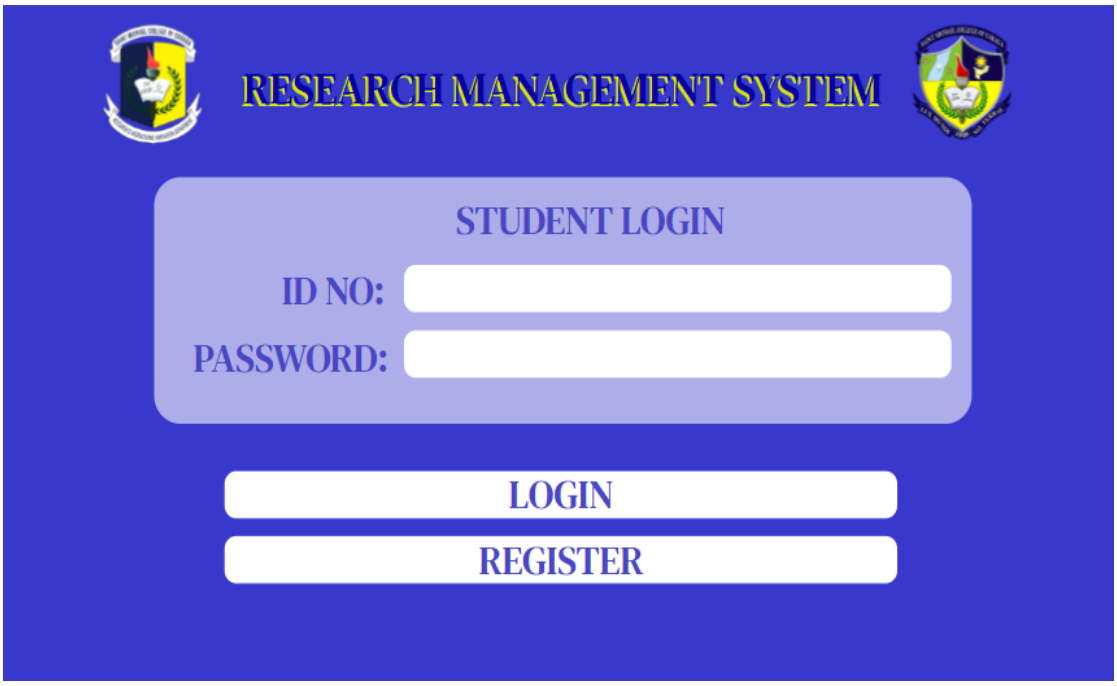


Figure 13: Login Page – Student

Figure #10 shows the login interface of the students. It consists of a textbox for their ID number and their password. In case they don't have their own accounts yet, they can register by clicking on the button below the login button labeled "REGISTER". After entering all the required credentials, they can try logging in using the login button. If the credentials are correct when trying to log in, the system will proceed to the account's dashboard.

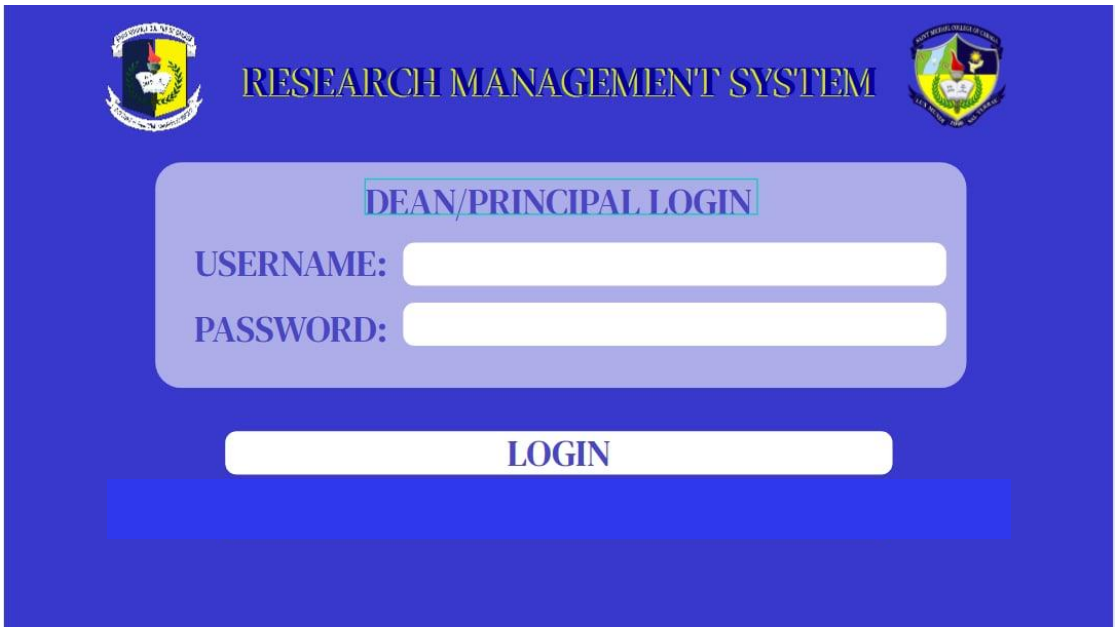


Figure 14: Login Page – Non-Students

Figure #11 shows the login page for Non-Student accounts (Adviser, Panel, Dean & Principal, Admin), which authorized users must complete to access the system. Depending on the account type, they can either manipulate, add, score, or view the contents of the research and the researchers' details. Dashboards are different but have a similar structure, where a table is the main component that composes different data (if there's any). Uniquely, external users won't be able to register an account for a Non-Student.



Figure 15: Student Dashboard – Title Justification

Note: This demonstration of the RMS will focus solely on the function of adding title justification and Gantt Chart display for simplicity, experimentation, and development of the algorithm. Other functions present in this User Interface are for visual purposes only.

The presented Figure is the landing page of the students after they log in. The user will be presented with the list of title justifications they have already submitted (if any), which can be edited or removed accordingly. If they want to add an entry, they can press the ADD button at the top-left of the table. A form will then appear *(see following Figure)* that they can fill out for the listing if the title needs justification. After adding, modifying, or removing an entry, the hybrid encryption algorithm will perform its process and store the encrypted data in the system. When in the dashboard, every title/entry would first be encrypted, meaning no data would be visible to the user immediately, including when inspecting the elements of the page. The entry will be visible if the user interacts with it.

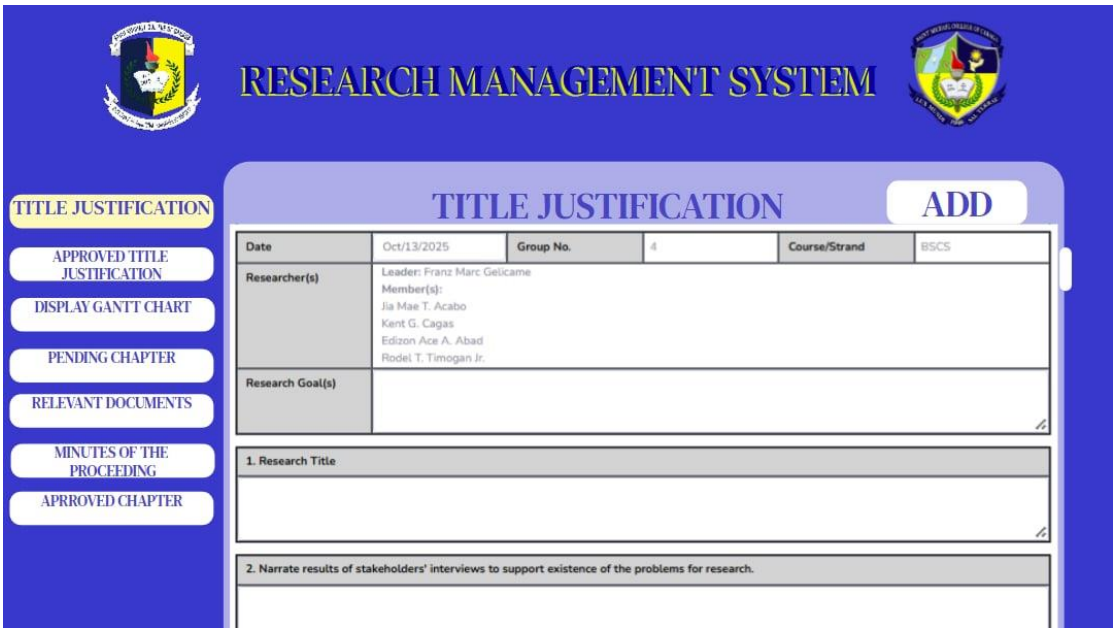


Figure 16: Student Dashboard – Title Justification Fill Out

Note: This demonstration of the RMS will focus solely on the function of adding title justification and Gantt Chart display for simplicity, experimentation, and development of the algorithm. Other functions present in this User Interface are for visual purposes only.

Figure # shows the form to fill out an entry for a title justification. The form will automatically fill out the field with data that is already present in the system (Student Name, Control No., Date, etc.). After filling it out, you can press add for final submission. In case of changes to the existing entry, you can edit it in the Title Justification Dashboard; at the same time, you can also remove it from your list.

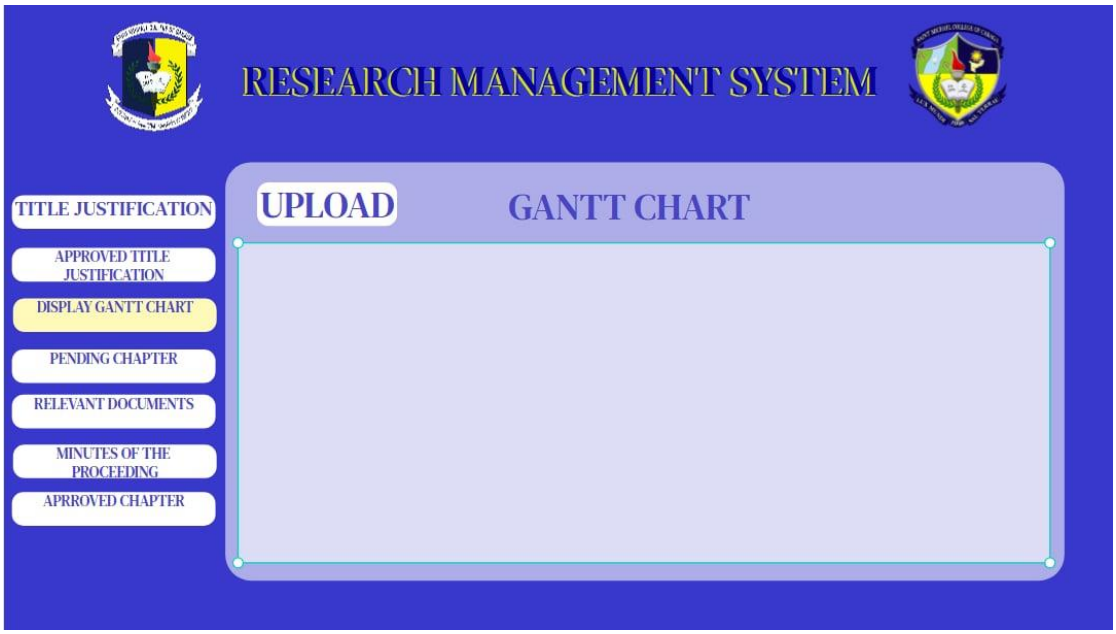


Figure 17: Student Dashboard – Gantt Chart

Note: This demonstration of the RMS will focus solely on the function of adding title justification and Gantt Chart display for simplicity, experimentation, and development of the algorithm. Other functions present in this User Interface are for visual purposes only.

Figure # shows the user interface when we move on to the “Display Gantt Chart” tab. This section will display your recently uploaded Gantt Chart that’ll be used by the Panelist, Advisers, and/or Dean/Principal for reference. The Gantt Chart will be signed personally, and the updated chart should always be uploaded to keep up-to-date. If there's no Gantt Chart yet, the user can add a blank or initially signed Gantt Chart to keep the Non-Student updated on their progress.

2.11 Software Platforms, Development Environments, and Tools

Table 1

Software Requirements

Category	Details
Front-End	**Platform:** Web-based (browser-accessible) **Languages:** HTML5, CSS3, JavaScript (ES6+) **Frameworks/Libraries:** React.js or Angular; Bootstrap/TailwindCSS **Tools:** Visual Studio Code (IDE), Git (version control)
Back-End	**Platform:** Server-side web application **Languages:** Python or Java **Frameworks:** Django (Python) / Spring Boot (Java) **Database:** MySQL or PostgreSQL **Encryption Libraries:** PyCryptodome (Python) / Java Cryptography Extension (JCE) **Tools:** Postman (API testing), Docker (containerization), GitHub/GitLab (repository)
Development Environment	**OS:** Windows 10/11 or Linux (Ubuntu) **IDE:** VS Code, PyCharm, IntelliJ IDEA **Deployment:** Apache or Nginx server; AWS/Azure cloud hosting

2.12 Hardware Requirements

Table 2

Hardware Requirements

Components	Specification	Usage
Device	RAM	A hardware component that is crucial to a system unit. It allows reading and writing of data in quick succession, and it serves as temporary storage. This component communicates directly to the CPU, which takes advantage of its swift operation. The RAM is volatile, meaning once it loses power, all the data it's holding will be lost.
	Central Processing Unit (CPU)/ Processor	The brain of the computer. This is where all the processing happens, thousands of instructions and computations being executed in just a fraction of a second. Very vital for the study as it is the one who will do the majority of the algorithm and would heavily rely on the efficiency it outputs.
	SSD/HDD	A component for data storage. It is a non-volatile hardware, meaning data will be kept even after turning off the system. This is necessary for the unit as it is where the RAM gets the data needed for the algorithm.

2.13 System Evaluation/Evaluation Procedure

The evaluation procedure for the proposed system "A Hybrid Encryption Algorithm Using AES and Blowfish for Institutional Systems" will follow a systematic process to assess its performance, security, and usability. The evaluation ensures that the system meets institutional standards for confidentiality, integrity, and availability.

Evaluation Objectives

The system evaluation aims to:

- Determine the effectiveness of the hybrid encryption algorithm in securing institutional data.
- Assess system usability, functionality, and reliability from end-users' perspectives.
- Measure efficiency and response time in data encryption and decryption.
- Validate compliance with ethical and security standards.

Evaluation Criteria and Instruments

Table 3

Evaluation Criteria and Instruments

Criteria	Description	Evaluation Method / Instrument
Functionality	Determines if all system features (encryption, decryption, user access, data handling) work as intended.	Functionality checklist, black-box testing
Performance	Assesses system speed, response time, and efficiency of encryption/decryption.	Benchmarking, stress testing, system logs
Security	Evaluates the confidentiality and integrity of data handled by the hybrid encryption	Penetration testing, vulnerability scanning
Reliability	Determines system stability and ability to handle errors or failures.	Fault testing, recovery testing
Maintainability	Evaluates ease of updates, debugging, and future enhancements.	Developer review, code inspection

2.14 Ethical Standards

The study entitled “A Hybrid Encryption Algorithm Using AES and Blowfish for Institutional Systems” was conducted for Saint Michael College of Caraga with the primary objective of enhancing data security against potential malicious threats. Throughout the conduct of the research, the researchers strictly adhered to established ethical principles and complied with Republic Act No. 10173, otherwise known as the Data Privacy Act of 2012, and its Implementing Rules and Regulations (IRR). All data collected, including user information and research-specific data, were handled with due diligence to ensure confidentiality, integrity, and availability, thereby preventing unauthorized access, disclosure, or misuse.

The institution and the administrators of the relevant systems and databases were duly informed regarding the purpose of the study and the manner in which the data were utilized within the proposed hybrid encryption algorithm. This study aimed to develop a justifiable, transparent, and user-friendly encryption solution while safeguarding the rights, privacy, and welfare of all stakeholders involved.

A. Intellectual Property Rights (IPR)

The researchers utilized established and widely recognized encryption algorithms, namely the Advanced Encryption Standard (AES) and Blowfish, which are publicly available and free from patent restrictions. Any open-source software or external resources used in the study were properly acknowledged in accordance with their respective licensing agreements.

All original concepts, implementations, and analyses developed during the research were clearly identified as the researchers’ own work, while appropriate citations were provided for borrowed ideas, references, or code segments. Copyright laws and intellectual property regulations were strictly observed, and no proprietary or restricted technologies were used without proper authorization.

B. Informed Consent

All participants involved in surveys, interviews, system testing, or evaluation activities were fully informed about the nature, objectives, and scope of the study. Participation was entirely voluntary, and informed consent was obtained prior to data collection. Participants were informed of their right to withdraw from the study at any stage without penalty or adverse consequences.

C. Data Privacy and Confidentiality

In compliance with Republic Act No. 10173 (Data Privacy Act of 2012), the researchers implemented appropriate organizational, physical, and technical security measures to protect personal and sensitive information. Identifiable personal data were anonymized, encrypted, and stored securely, and no personally identifiable information was disclosed in the research outputs. All collected data were used solely for academic and research purposes related to this study.

D. Voluntary Participation

Participation in the study was strictly voluntary. No individual was coerced or unduly influenced to participate, and participants were assured that their decision to participate or withdraw would not affect their academic or professional standing within the institution.

E. Minimization of Harm

The researchers ensured that the study posed no physical, psychological, social, or reputational risks to participants or the institution. System testing and evaluation were conducted in a controlled environment to prevent disruption, data loss, or security breaches within the institutional systems.

F. Beneficence

The study was designed to contribute positively to the institution by strengthening data protection mechanisms and reducing vulnerabilities to cyber threats. The proposed hybrid encryption algorithm is intended to benefit the institution, its administrators, and users by enhancing the overall security and reliability of institutional data.

G. Justice and Fair Participant Selection

Participants and data sets were selected based solely on their relevance to the research objectives (personal information, files, and varying data based on their size). No form of discrimination based on personal, academic, or social attributes influenced participant selection. All contributions were treated equitably and respectfully.

H. Data Integrity and Accuracy

The researchers committed to maintaining the accuracy, reliability, and integrity of all data collected and analyzed through the implementation of systematic verification and validation procedures. Data accuracy was ensured by performing cross-checks, consistency checks, and controlled testing prior to analysis. Additionally, system-generated audit logs were utilized to record data access, modifications, and processing activities, enabling traceability and accountability throughout the research process. All datasets underwent periodic review and verification to detect errors, unauthorized changes, or inconsistencies, ensuring that the reported results were accurate, unbiased, and free from manipulation or misrepresentation.

I. Transparency in Reporting

The findings of the study were reported honestly and comprehensively, including both successful outcomes and identified limitations. The researchers maintained openness to peer review, validation, and constructive feedback to uphold academic integrity and accountability.

J. Use of Patented or Copyrighted Materials

Any patented, copyrighted, or proprietary software tools used during the study were properly cited and utilized strictly for educational and research purposes. Necessary permissions and licenses were secured when required, and no materials were used for commercial gain.

K. Ethical Considerations for Animal and Human Trials

No animal subjects were involved in this research. All interactions with human participants adhered to ethical research standards, including informed consent, confidentiality, and voluntary participation, consistent with institutional guidelines and the Data Privacy Act of 2012.

L. Responsible use of AI and Related Technologies

Artificial intelligence and related computational tools, when used, were employed responsibly and ethically. These technologies were utilized primarily for data organization, analysis, documentation, and research assistance, without compromising participant privacy, data security, or ethical standards. All AI-assisted processes complied with applicable data protection and research ethics guidelines.

M. Ethical Approval

The research project obtained ethical clearance from the designated Ethics Committee Representative of Saint Michael College of Caraga prior to its implementation. This approval confirmed that the study complied with institutional ethical standards, data privacy regulations under Republic Act No. 10173, and established principles governing intellectual property, participant rights, and data confidentiality. All research activities were conducted responsibly and in accordance with approved ethical protocols.

4.4 References

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APPENDICES

A - Letter of Approval with Certificate of Appearance and Photo
B - Instrument with Informed Consent
C - Map of the Research Locale *(Include Coordinates)*
D - Curriculum Vitae
E - Narrative & Photo Documentation
F- Ethics Committee Approval

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